Rs 120/-

MULTIMEDIA

ELECTIVE(III)

BCT

CHAPTER WISE NOTE

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COMPLETE

OLD QUESTION SOLUTION

(Reference Prof. Dr. Subarna sir slide)

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2072 BATCH

BEST OF LUCK FOR THE EXAM

Unit 1. Introduction: Multimedia

Q. What is Multimedia?

- Multimedia means that computer information can be represented through audio, video, and animation in addition to traditional media (i.e., text, graphics/drawings, images).

General Definition

A good general working definition for this module is:

Multimedia is the field concerned with the **computer-controlled** integration of text, graphics, drawings, still and moving images (Video), animation, audio, and any other media where every type of information can be represented, stored, transmitted and processed **digitally**.

ELEMENTS OF MULTIMEDIA

Text

Text is the foundation for word processing programs and is still the fundamental information used in many multimedia programs. In fact, many multimedia applications are based on the conversion of a book to a computerized form.

Static Graphics images:

When you imagine graphics images you believably think of "still" images-that is, images such as those in a photograph or drawing. There is no occurrence in these kinds of picture. Still graphics images are an all-important portion of multimedia because humans are modality adjusted.

Animation:

Animation mention to moving graphics images. The happening of somebody giving CPR makes it much easier to learn internal organ revitalization, rather than just screening a static picture.

Full-Motion Video:

Full-motion video, such as the images depicted in a television, can add even more than to a multimedia application. Although full-motion video may sound similar a perfect way to add a powerful message to a multimedia application, it is nowhere near the quality you would anticipate after watching television.

Audio Sound:

The combination of audio sound into a multimedia application can offer the user with information not likely finished any other technique of announcement. Some types of information can't be taken efficiently without using sound.

Multimedia Application:

- A Multimedia Application is an application which uses a collection of multiple media sources e.g. text, graphics, images, sound/audio, animation and/or video.

USES AND APPLICATIONS OF MULTIMEDIA:

- Multimedia in Education

Scanned with CamScanner

- Multimedia in Games
- Multimedia in Training
- Creative industries
  Creative industries use multimedia for a variety of purposes ranging from fine arts, entertainment, to commercial art, to journalism, to media and software servi provided for any of the industries
- Science and Technology
- Language communication
- Medicine:
  In medicine, doctors can get trained by looking at a virtual surgery or they c simulate how the human body is affected by diseases spread by viruses and bacte and then develop techniques to prevent it.

FEATURES OF MULTIMEDIA:

Photo gallery:

Show your photos arranged in a nice-looking grid format.

Slideshows:

Combine your pictures with music and animate them in a slideshow.

Audio player:

Add music, podcasts, or other audio files to your website.

Video player:

Upload videos and display them in a professional player or embed videos directly from v sharing websites such as http://www.youtube.com.

Embedded documents:

Embed already existing documents from script.com or other document sharing web directly into your website for easy viewing.

ADVANTAGES OF MULTIMEDIA

Creativity:

It brings more life to discussions.

Variety:

It caters all types of learners.

Cost-effective:

Multimedia mostly requires only a one-time purchase of devices and software, which c used unlimited times thereafter.

Evaluation:

It offers ideal learning assessment tools which are also entertaining for the students.

Realistic Approach:

It provides approaches which make learning more realistic.

Wide Variety of Support:

Multiple media formats are available for use, with different models being able to multimedia

Trendy:

The current trend of culture leans toward technology, and a great number of resources are being made available for different media formats.

DISADVANTAGES OF MULTIMEDIA

Accessibility:

Multimedia definite quantity electricity to be operated, which may not be accessible in some rural areas or may not be systematically acquirable due to deficit and intermission.

Distracting:

Multimedia may take away the focusing from the lesson due to its attention-grabbing formats.

Costly:

Production of multimedia is more costly than others because it is made up of more than one medium. Production of multimedia requires an electronic device, which may be relatively expensive. Multimedia requires electricity to run, which adds to the cost of its use.

Time Consuming:

Creating multimedia requires more time.

Requires Mastery:

Multimedia requires consistent and long practice to master, which may take a lot of time and energy from the user.

Limited Support/Compatibility:

There is a wide variety of gadget models which arouses incompatibilities of media formats.

Fragile:

The device used for multimedia must be used with attention; vulnerability to moisture or other elements could cause costly, reparable damage which would require another purchase of a device.

Q. What is Hypertext and Hypermedia?

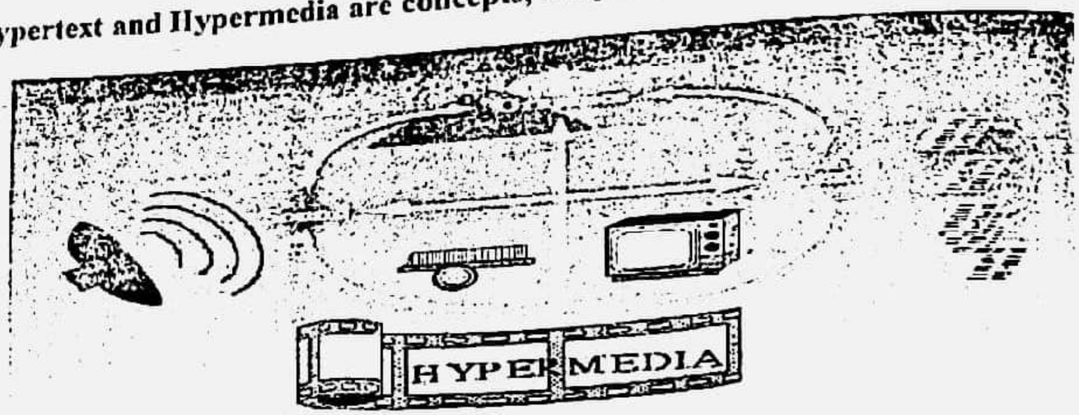
Hyper-Text:

- Hypertext is text which is not constrained to be linear.
- Hypertext refers to a word, phrase or chunk of text that can be linked to another document or text.
- Hypertext covers both textual hyperlinks and graphical ones.
- The term was coined by Ted Nelson in the 1960s and is one of the key concepts that makes the Internet work. Without hypertext, following a link on a topic to a related article on that topic – one of the primary means of navigating the Web – would be impossible.

Hypermedia

- Hypermedia is a term used for hypertext which is not constrained to be text: it can include graphics, video and sound, for example. Apparently, Ted Nelson was the first to use this term too.

Hypertext and Hypermedia are concepts, not products.



- The World Wide Web (WWW) is the best example of a hypermedia application.
- Adobe Acrobat
- PowerPoint

Multimedia Systems

- A Multimedia System is a system capable of processing multimedia data and applications
- A Multimedia System is characterized by the processing, storage, generation, manipulation of Multimedia information.
- A computer system with the capabilities to capture, digitize, compress, store, decompress and present information is called multimedia system.
- The aim of multimedia system is to provide a creative and effective way of producing, storing and communicating information.

Characteristics of a Multimedia System

A Multimedia system has four basic characteristics:

- Multimedia systems must be **computer controlled.**
- Multimedia systems are **integrated.**
- The information they handle must be represented **digitally.**
- The interface to the final presentation of media is usually **interactive.**

Challenges for Multimedia Systems

- Render different data at same time - continuously.
- Sequencing within the media
- Playing frames in correct order/time frame in video
- Synchronization — inter-media scheduling

Key Issues for Multimedia Systems

- How to strictly maintain the temporal relationships on play back/retrieval
- What process are involved in the above.
- Data has to represented digitally — Analog–Digital Conversion, Sampling etc.
- Large Data Requirements — bandwidth, storage, compression

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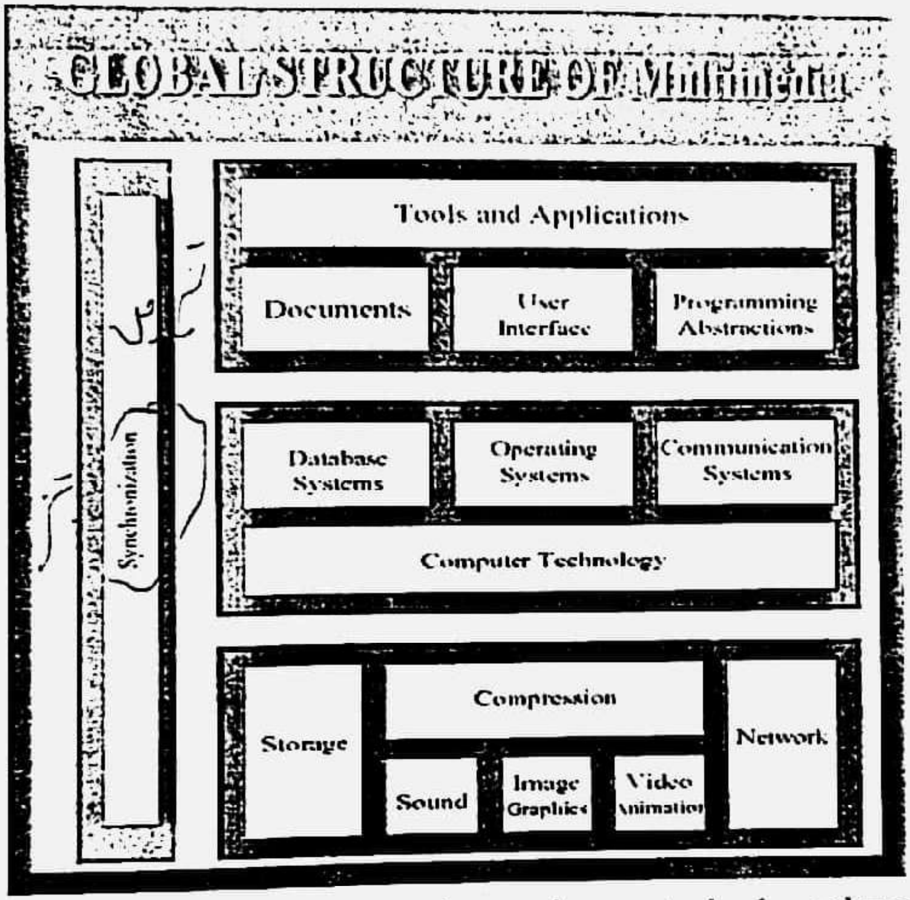
Global Structure



Application domain — provides functions to the user to develop and present multimedia projects. This includes Software tools, and multimedia projects development methodology.
System domain — including all supports for using the functions of the device domain, e.g. operating systems, communication systems (networking) and database systems.
Device domain — basic concepts and skill for processing various multimedia elements and for handling physical device.

Unit 2. Sound and Audio System

The Nature of Sound

- Sound is a physical phenomenon produced by the vibration of matter and transmitted as waves. However, the perception of sound by human beings Is a very complex process. It involves three systems:
- The source which emits sound;
- The medium through which the sound propagates;
- The detector which receives and interprets the sound.

Sounds we heard everyday are very complex. Every sound is comprised of waves of many different frequencies and shapes. But the simplest sound we can hear is a sine wave.
Sound waves can be characterized by the following attributes:
Period, Frequency (Pitch), Amplitude (Loudness) Bandwidth (Dynamic)

1.1 Pitch and Frequency

- Period is the interval at which a periodic signal repeat regularly.
- Pitch is a perception of sound by human beings It measures how 'high' is the sound as itis perceived by a listener.
- Frequency measures a physical property of a wave. It is the reciprocal value of period f= 1/p the unit is Herts (Hz) or kilohertz (kHz).

Musical instruments are tuned to produce a set of fixed pitches.
Infra-sound 0—20 Hz
Human hearing range 20 —20 kHz
Ultrasound 20 kHz— 1 GHz
Hyper sound 1 GHz — 10 THz

1.2. Loudness and Amplitude

- The other important perceptual quality is loudness or volume.
- Amplitude is the measure of sound levels. For a digital sound, amplitude is the sample value.
- The reason that sounds have different loudness is that they carry different amount of power. The unit of power is watt. The intensity of sound is the amount of power transmitted through an area of $1m^2$ oriented perpendicular to the propagation direction of the sound.
- If the intensity of a sound is 1watt/m^2, we may start feel the sound. The ear may be damaged. This is known as the threshold of feeling. If the intensity is 10 "watt /m^2, we may just be able to hear it. This is known as the threshold of hearing.

The relative intensity of two different sounds ismeasured using the deciBel (dB). It is defined byrelative intensity in dB = 10 log I_2/I_1
Very often, we will compare a sound with the threshold of hearing.

1.3. Dynamic and Bandwidth

- Dynamic range means the change in sound levels.

For example, a large orchestra can reach 130dB at its climax and drop to as low as 30dB at its softest, giving a range of 100dB.

- Bandwidth is the range of frequencies a device can produce or a human can hear.

FM radio 50Hz — 15kHz
AM radio 80Hz — 5kHz
CD player 20Hz — 20kHz
Sound Blaster 16 soundcard 30Hz— 20kHz
Inexpensive microphone 80Hz — 12kHz
Telephone 300Hz — 3kHz
Children's ears 20Hz — 20kHz
Older ears 50Hz — 10kHz Male voice 120Hz — 7kHz
Female voice 200Hz — 9kHz

2 Computer Representation of Sound

- Sound waves are continuous while computers are good at handling discrete numbers.
- In order to store a sound wave 1n a computer, samples of the wave are taken.
- Each sample is represented by a number, the 'code'.
- This process is known as digitization.
- This method of digitizing sound is known as pulse code modulation (PCM). Refer to Unit 1 for more information on digitization.
- According to Nyquist sampling theorem, 1n order to capture all audible frequency components of a sound, i.e., up to 20h. Hz, we need to set the sampling to at least twice of this. This is why one of the most popular sampling rates for high quality sound is 4410 Hz.
- Another aspect we need to consider 1s the resolution, i.e., the number of bits used to represent a sample.
- Often, 16 bits are used for each sample in high quality sound. This gives the SNR of 96d.

2.1 Quality versus File Size

The size of a digital recording depends onthe sampling rate, resolution and number of channels.

S=Rx (b/8) *C* D

Where S= file size(bytes), R= Sampling Rates (samples per second), b = resolution, C = channels, D = Recording Duration

Higher sampling rate, higher resolution gives higher quality but bigger file size.

For example, if we record 10 seconds of stereo music at 44 1kHz, 16 bits, the size will be:

S = 44100 x (16/8) x 2x 10

= 1,764,000bytes

= 1722.7Kbytes

High quality sound files are very big; however, the file size can be reduced by compr

2.2 Audio File Formats

- The most commonly used digital sound format in Windows systems is .wav fil
- Sound is stored in .wav as digital samples known as Pulse Code Modulation (P
- Each .wav file has a header containing information of the file.

There is usually no compression in .wav files.

Other format may use different compression technique to reduce file size.

. Vox use Adaptive Delta Pulse Code Modulation (ADPCM). Eg mp3 MPEG-1-layer 3

RealAudio file is a proprietary format.

2.3. Audio Hardware

Recording and Digitizing sound:

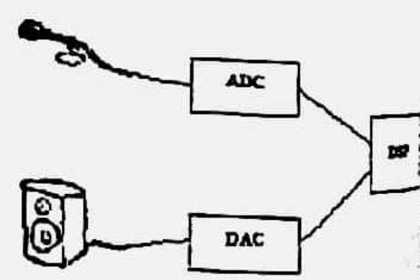
- An analog-to-digital converter (ADC) converts the analog sound signal into samples.
- A digital signal processor (DSP) processes the sample, e.g. filtering, mod compression, and so on.

Play back sound:

- A digital signal processor processes the sample, e.g. decompression, demod and so on
- A digital-to-analog converter (DAC) converts the digital samples into sound s
- All these hardware devices are integrated into a few chips on a sound card

Different sound card has different capability of processing digital sounds. When b sound card, you should look at:

- maximum sampling rate « stereo or mono
- duplex or simplex
- stereo or mono



2.4 Audio Software

Windows device driver — controls the hardware device.

Many popular sound cards are Plus and Play. Windows has drivers for them recognize them automatically. For cards that Windows does not have drivers, you ne the driver from the manufacturer and install it with the card.

If you do not hear sound, you should check the settings, such as interrupt, DMA chan so on.

Device manager — the user interface to the hardware for configuring the devices.

You can choose which audio device you want to use

You can set the audio volume

Mixer — its functions are:

Unit 3. Images and Graphics

1. The Nature of Digital Images

- An image is a spatial representation of an object, a two-dimensional or three-dimensional scene or another image. Often the images reflect the intensity of lights.
- Most photographs are called continuous-tone images because the method used to develop the photograph creates the illusion of perfect continuous tone throughout the image.
- Images stored and processed by computers, displayed on computer screens, are called digital images although they often look like continuous-tone. This is because they are represented by a matrix of numeric values each represents a quantized intensity value.

1.1 Basic Concepts

The smallest element on a digital image is known as a pixel —a picture element. A digital image consists of a (usually rectangular) matrix of pixels.

1.2 Depth

The depth of an image is the number of bits used to represent each pixel.

1-bit black-and-white image, also called bitmap image.

4-bit can represent 16 colours, used in low resolution screens (EGA/VGA)

8-bit can have 256 colours. The 256 colourimages are often known as indexed colour images. The values are actually indexing to a table of many more different colors. For example, Color 3 is mapped to (200. 10, 10).

8-bit grey 256 prey-levels. The image containsonly brightness/intensity data without colour information.

16-bitcan have 65536 colours, also known as hi-colour in Windows systems. The 16 bits are divided into 5 bits for RED, 6 bits for GREEN and 5 bits for BLUE.

24-bit — 2^{24}= 16. 777,216 colours, true colour. Each byte is used to represent the intensity of a primary colour. RED, GREEN and BLUE. Each colour can have 256 different levels.

1.3. Resolution

- Resolution measures how much detail an image can have. There are several resolutions relating toimages.

Image resolution is the number of pixels in an image.

320 x 240 = 76800 pixels, 700 x 400 = 280000 pixels

Display (Monitor) resolution — refers to number of dots per inch (dpi) on a monitor. Windows systems usually have 96dpi resolution. Some high-resolution video adapters/monitors support 120dpi. For example, 288 * 216 images displayed on a monitor with 96dpi will be 3" x 2*1/4",

Output resolution — refers to number of dots per mech (dpi) on a (hard copy) output device. Many printers have 300dpi or 600 dpi resolution. High-quality imagesetters can print at a range between 1200dpi and 2400dpi, or higher. The above image printed on a 300dpi printer will be 0.96 * 0.72 inch.

1.4 Acquiring Digital Images

There are many ways to create or get digital images. We list some of the most common ways:

- to combine sound from different sources
- to adjust the play back volume of sound sources
- to adjust the recording volume of sound sources

Recording — Windows has a simple Sound Recorder.
Editing — The Windows Sound Recorder has a limiting editing function, such as chang volume and speed, deleting part of the sound.
There are many freeware and shareware programs for sound recording, editing processing.

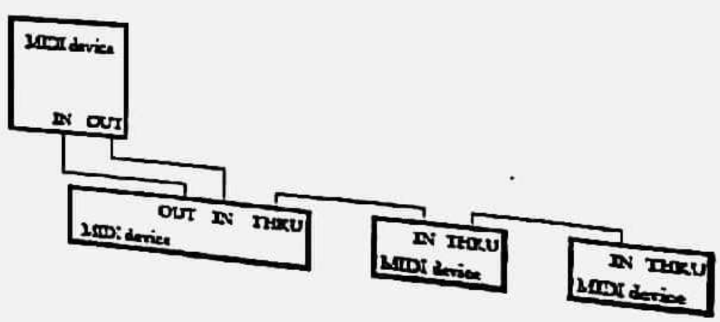
3 Computer Music — MIDI

- Sound waves, whether occurred natural or man-made, are often very complex, they consist of many frequencies. Digital sound is relatively straight forward to rec complex sound. However, it is quite difficult to generate (or synthesize) comp sound.
- There is a better way to generate high quality music. This is known as MIDI Musical Instrument Digital Interface.
- It is a communication standard developed in the early 1980s for electronic instrum and computers. It specifies the hardware connection between equipment's as well the format in which the data are transferred between the equipment's.
- Common MIDI devices include electronic music synthesizers, modules, and MI devices in common sound cards.
- General MIDI is a standard specified by MIDI Manufacturers Association. To be G compatible, a sound generating device must meet the General MIDI system leve performance requirement.

3.1 MIDI Hardware

- An electronic musical instrument or a computer which has MIDI interface should one or more MIDI ports. The MIDI ports on musical instruments are usually lab with:

IN — for receiving, MIDI data;
OUT — for outputting MIDI data that are generated by the instrument:
THRU — for passing MIDI data to the next instrument.



- Make an image from scratch with a paint program. A good program will allow you to choose the depth, resolution and size.
- Grab an image of a screen. The depth, resolution and size is determined by the screen.
- Capture an image from a digital camera or a camcorder. The depth, resolution and size is determined by the camera or the camcorder. The popular depth 1s 24-bit. The commonly used resolution 1s 320) x 240, 640 x 480 and 800 x GOD.
- Scan a photograph or a print using a scanner. You can select from a range of different depths and resolution. The choice should be determined by the type of original and the final output form.
- Convert from existing digital media — e.g., photoCD. The attribute is determined by the original image.
- Synthesize an image from numerical data.

2 Vector Graphics

Instead of using pixels, objects can be represented by their attributes, such as size, colour, location, and so on. This type of graphics is known as vector graphics, or vector drawing. This is an abstract representation of a 2-dimensional or 3-dimensional scene.

A vector graphics file contains graphics primitives, for example, rectangles, circles, lines.

There are many languages for describing vector graphics. Three of them are very popular. They are:

PostScript was developed by Adobe as a page description language. The next page shows a graphic with its PostScript program source.

VRML stands for Virtual Reality Markup Language. It is for descripting a scene in a virtual world.

SVG stands for Scalable Vector Graphic. It is a language for describing two-dimensional graphicsin XML. It allows three types of graphics objects: vector graphic shapes, images and text.

2.1 Vector versus Bitmap

Vector	**Bitmap**
A vector graphic contains mathematical description of objects	A bitmap contains an exact pixel-by-pixel value of an image
A vector graphic 1s resolution independent	A bitmap file is fixed in resolution
The file size of a vector graphic depends on the number of graphic elements it contains	The file size of a bitmap is completely determined by the image resolution and its depth.
Displaying a vector graphic usually involves a large amount of processing	A bitmap image is easier to render

3 Colour Systems

Colour is a vital component of multimedia. Colour management is both a subjective and a technical exercise, because:

- Colour is a physical property of light, but Colour perception is a human physiologic activity. Choosing a right colour or colour combination involves many trials & aesthetic judgement.
- Colour is the frequency/wave-length of a light wave within the narrow band of theelectromagnetic spectrum (380 — 760nm) to which the human eye responds.

3.1 RGB Colour Model

This is probably the most popular colour model used in computer graphics.
It is an additive system in which varying amount of the three primary colours, red, green and blue, are added to black to produce new colours.
You can imagine three light sources of the primary colours shine on a black surface. By varying the intensity of the lights, you will produce different colours.
R — Red
G — Green
B — Blue

3.2 CMY Colour Model

This model is based on the light absorbing quality of inks printed on paper. Combining three primary colour pigments, Cyan, Magenta and Yellow, should absorb all light, thus resulting in black.
It is a subtractive model.
The value of each primary colour is assigned a percentage from the lightest (0%) to the darkest (100%).
Because all inks contain some impurities, three inks actually produce a muddy brown, a black colour is added in printing process, thus CMYK model.
Note: the primary colours in RGB and CMY models are complementary colours.
C— Cyan
M — Magenta
Y — Yellow

3.3. HSB Colour Model

This model is based on the human perception of colour.
The three fundamental characteristics of colours are:
Hue — is the wavelength of the light. Hue is often identified by the name of the colour. It is measured as a location on the standard colour wheel as a degree between 0 to 360.
Saturation — is the strength or purity of the colour. It represents the amount of gray in proportion to the hue and is measured as a percentage from 0°%(gray) to 100% (fully saturated).
Brightness — is the relative lightness or
darkness of the colour. Itis measured as a percentage from 0%o(black) to 100%(whute).

3.4 YUV Colour Model

This model is widely used in encoding colour for use in television and video.
The theory behind this model is that human perception is more sensitive to brightness than any chrominance information, so a more suitable coding distinguishes between luminance and

chrominance. This also produces a system that is compatible with black-and-white TV systems.
The Y-signal encodes the brightness information. Black-and-white television system will use this channel only.
The U and V channels encode the chromatic information. The resolution of the U and V channels is often less than the Y channel for the reason of reducing the size.

3.5 Gamut

The gamut of a colour system is the range of colours that can be displayed or printed. The spectrum of colours that be viewed by human eye is wider than any method of reproducing colour.
Different colour models have different gamut. The CMYK model is smaller than RGB model. On the right is a Chromaticity Diagram which illustrates gumat of RGB and CMYK colour systems.

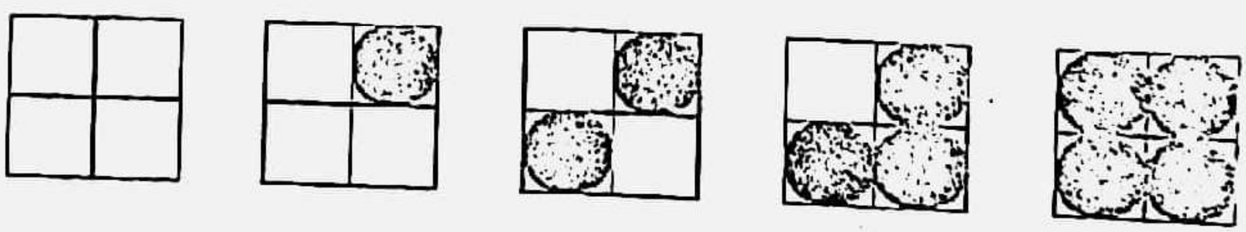
3.6 Colour Palette

A colour palette is an index table to available colours in an indexed colour system. When working in 8-bit mode, a system can display only 256 colours out of a total of 16 million colours. The system keeps a default palette of available colours.
Palette flashing. Each program may have its own palette. It may replace the system palette with its own for the period it is active. This may cause an annoying flash of strange colours in your screen, known as palette flashing. This is a serious problem in multimedia applications.

4 Some Image Techniques

4.1 Dithering

Dithering is a technique to increase the number of colours to be perceived in an image. It is based on human eye's capability for spatial integration, that is, 1f you look at a number of closely placed small objects from a distance, they will look like merged together.
Dithering technique groups a number of pixels together, say 4, to form a cluster. When viewed from sufficient distance, the individual pixel will not be distinguishable. The cluster will look like a single block of a colour different from the individual pixel



5 Image and Graphics File Formats

A digital image 1s stored in a file conforming to certain format. In addition to the pixel data, the file contains information to identify and decode the data:

- The format
- The image sizes
- Depth
- Colour and palette
- Compression

Some formats are defined to work only in certain platform while other can be used for platforms. Some formats are specific for an application. Some formats are for images, others are for vector graphics. Some formats allow compression, others contain only raw data.
Note: Formats using compression will make the file size smaller. Some compression algorithms will lose some image information.

6 Digital Image Processing

This is a very large area contain the following sub-areas:
Image analysis is concerned with techniques for extracting descriptions from images that are necessary for higher-level scene analysis methods.
Image recognition is concerned with the techniques for recovering information about objects im the image. A sub-area 1s character recognition.
image enhancement is concemed with the technique to improve the image and to correct some defects, such as

- colour and tonal adjustment,
- Transformations, eg scale, rotate,
- Special effects, eg texture, stylize, blur, sharpen.

Unit: 4 Video and Animation

Motion

- Both video and animation give us a sense of motion
- They exploit some properties of human eye's ability of viewing pictures
- Motion video is the element of multimedia that can hold the interest of viewers in a presentation.

Visual Representation

- The visual effect of motion is due to a biological phenomenon known as persistence of vision
- An object seen by the human eye remains mapped on the eye's retina for a brief time after viewing (approximately 25 ms)
- Another phenomenon contributing to the vision of motion is known as phi phenomenon
- When two light sources are close by and they are illuminated in quick succession, what we see is not two lights but a single light moving between the two points
- Due to the above two phenomena of our vision system, a discrete sequence of individual pictures can be perceived as a continuous sequence
- Temporal aspect of Illumination—To represent
- visual reality, two conditions must be met the rate of repetition of the images must be high enough to guarantee smooth motion from frame to frame.
- The rate must be high enough so that the persistence of vision extends over the interval between flashes

- The frequency at which the flicking light source must be repeated before it appears continuous is known as the fusion frequency
- This depends on the brightness of the light source
- The brighter the light source the higher the fusion frequency

- It is known that we perceive a continuous motion to happen at any frame rate faster than 15 frames per second
- PAL television system has a frame rate of 25 frames/s
- Another problem known as flicker occurs due to a periodic fluctuation of brightness perception
- A technique known as interleaving improves the view by
  - dividing a frame into two fields, each contains the alternative scan lines, and
  - displaying the field in twice of the frames rate

Videoresolution

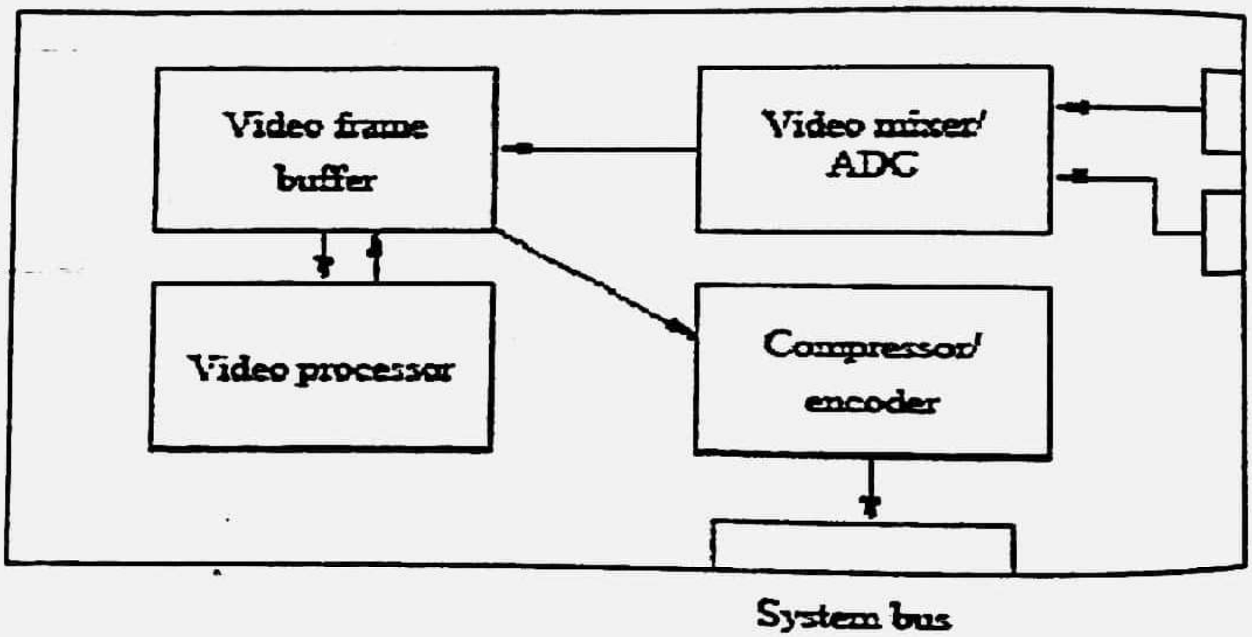
- The smallest detail that can be reproduced in the image is a pixel
- Practically, some of the scene inevitably fall between scanning lines, so that two lines are required for such picture elements
- Only about 70% of the vertical detail is presented by the scanning lines
- Aspect ratio is the ratio of the picture width to height.
- It is 4:3 for conventional TV
- The picture width, horizontal resolution and the total detail content of the image can be calculated
- Conventional video systems have relative low resolution
  - o compare to computer screens: typical resolution of
- 640 x 480, even up to 1024 X 768
- One consequence of this low resolution is that video played on computer screen are usually in a small window
- On the other hand, even with this low resolution, the amount of data in video is huge
- Consider PAL TV at 25 frames per second, if we sample at 352 x 288 with 16 bits per pixel, the raw video size is
- 352 x288 x16 x 25 = 40.55Mbit/s = 5Mbytes/s
- **Digitalizing Video**
- We need to capture or digitize video for playing back on computers or integrating into multimedia applications
- We need to take a lot of samples
- At 25 frames per second, each frame requires 1/25 = 40ms
- There 625 scan lines in each frame, giving each scan line is40ms/625 = 64μs
- At a horizontal resolution of 425 pixel, the time for sampling each pixel is 64μs/425 = 0.15μsi.e., sampling rate is at least 7Mhz
- This requires very fast hardware
- Hardware required to capture video:

- Video sources: TV, VCR, Laserdisc player, Camcorder
- Video capture card
- Storage space: large hard disk

Video capturecards

There are many different video capture cards on the market the common features in these cards are:

- Can accept composite video or S-VHS in NTSC or PAL; high-end capture cards can accept digital video (DV)
- Video input mixer and ADC— to select/combine video sources, to convert analog video signal to digital samples
- Video frame buffer— temporary storage for video frame
- Video processor— to filter or enhance the video frame, e.g., reduce noise, adjust brightness, contrast and colour
- Compressor/encoder— to compress and encode the digital video into a required format
- Interface to the system PCI bus



Video Format

- AVI (Audio Video Interleaved) format was defined by Microsoft for its Video f Windows systems
- It supports video playback at up to 30 frames per second on a small window (typic size 300X200 with 8 or 16-bit colour)
- It is a software-only system
- It supports a number of compression algorithms

- QuickTime was originally developed by Apple for storing audio and video in Macintosh systems
- It supports video playback at up to 30 frames per second on a small window (typical size 300X200 with 8 or 16-bit colour)
- It is a software-only system
- It supports a number of compression algorithms

Animation

- To animate something is, literally, to bring it to life
- An animation covers all changes that have a visual effect
- Visual effect can be of two major kinds:
- motion dynamic— time varying positions
- update dynamic— time varying shape, colour,
- texture, or even lighting, camera position, etc.
- The visual effects are the result of exploiting the properties of human vision system as described above (in the section about video)
- A computer animation is an animation performed by a computer using graphical tools to provide visual effects

UNIT 5: DATA COMPRESSION

Why Compress?

- To reduce the volume of data to be transmitted (text, fax, images).
- To reduce the bandwidth required for transmission and to reduce storage requirements (speech, audio, video)

Data compression implies sending or storing a smaller number of bits. Although many methods are used for this purpose, in general these methods can be divided into two broad categories: **lossless and lossy methods.**

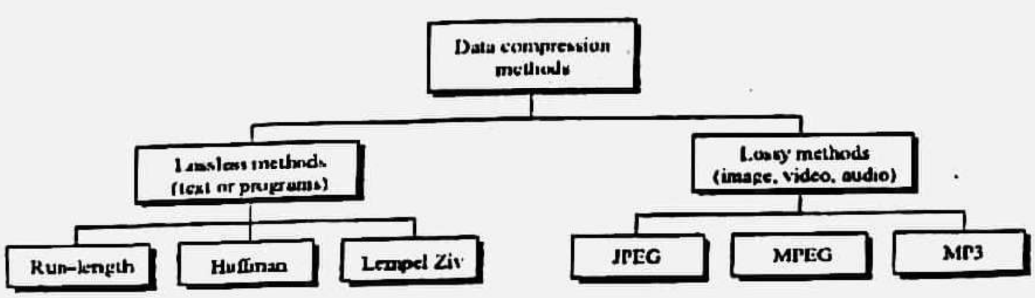


Figure1 Data compression methods

Compression

How is compression possible?

- Redundancy in digital audio, image, and video data
- Properties of human perception

1) Digital audio is a series of sample values; image is a rectangular array of pixel values; video is a sequence of images played out at a certain rate
2) Neighboring sample values are correlated.

Redundancy

- Adjacent audio samples are similar (predictive encoding); samples corresponding to silence (silence removal)
- In digital image, neighboring samples on a scanning line are normally similar (spatial redundancy)
- In digital video, in addition to spatial redundancy, neighboring images in a video sequence may be similar (temporal redundancy)

Human Perception on Factors?

- Compressed version of digital audio, image, video need not represent the original information exactly
- Perception sensitivities are different for different signal patterns
- Human eye is less sensitive to the higher spatial frequency components than the lower frequencies (transform coding)

Classification

1) **Lossless compression**
   - lossless compression for legal and medical documents, computer programs
   - exploit only data redundancy

2) **Lossy compression**
   - digital audio, image, video where some errors or loss can be tolerated
   - exploit both data redundancy and human
   - perception properties
   - Constant bit rate versus variable bit rate coding

Entropy

- Amount of information I in a symbol of occurring probability p : $I = \log2(1/p)$
- Symbols that occur rarely convey a large amount of information
- Average information per symbol is called entropy H

$$H = p_i x \log_2(1/p_i) \text{ bits per codeword}$$

- Average number of bits per codeword $=N_i p_i$
  where N_i is the number of bits for the symbol generated by the encoding algorithm

Huffman Coding

- Assigns fewer bits to symbols that appear more often and more bits to the symbols that appear less often
- Efficient when occurrence probabilities vary widely
- Huffman codebook from the set of symbols and their occurring probabilities

  Two Properties:
  - generate compact codes
  - prefix property

Run length coding

- Repeated occurrence of the same character is called a run
- Number of repetitions is called the length of the run
- Run of any length is represented by three characters
  - eeeeeee7tnnnnnnnn
  - @e7t@n8

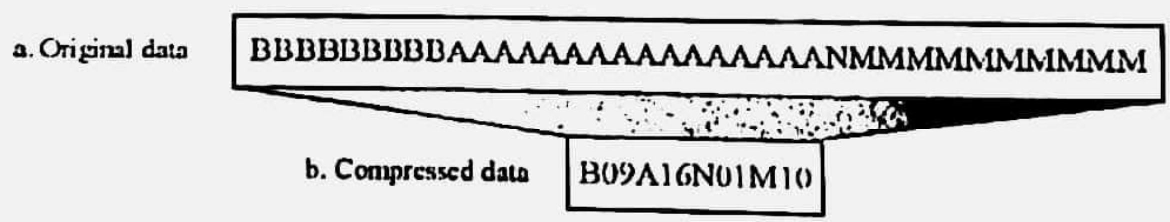


Figure 15.2: Run-length encoding example

Lempel.Zi.Welch(Lzw) Coding

- Works by building a dictionary of phrases from the input stream
- A token or an index is used to identify each distinct phrase
- Number of entries in the dictionary determines the number of bits required for the index -- a dictionary with 25,000 words requires 15 bits to encode the index.

Arithmetic Coding

String of characters with occurrence probabilities make up a message
A complete message may be fragmented into multiple smaller strings
A codeword corresponding to each string is found separately

Data Compression

Coding Requirements

Let us consider the general requirements imposed on most multimedia systems:

Storage — multimedia elements require much more storage space than simple text. For example, a full screen true colour image is 640 x 480 X 3 = 921600 bytes

The size of one second of uncompressed CD quality stereo audio is 44.1kHz X 2 X 2 = 176400 bytes

The size of one second of uncompressed PAL video is

$$384*288*3*25 \text{ frames} = 8294400 \text{ bytes}$$

Throughput — continuous media require very large throughput.

For example, an uncompressed CD quality stereo audio stream needs 176400 bytes/sec. A raw digitized PAL TV signal needs(13.5MHz + 6.75MHz + 6.75MHz) X 8bits

$$= 216 \times 10^6 \text{bits/sec}$$
$$= 27 \times 10^6 \text{Bytes/sec}$$

Interaction — to support fast interaction, the end-to-end delay should be small. A 'face-to-face' application, such as video conferencing, requires the delay to be less than 50ms. Furthermore, multimedia elements have to be accessed randomly.

Conclusion:

Multimedia elements are **very large**.

We need to reduce the data size using **compression**.

Kinds of coding methods

Lossless — the compression process does not reduce the amount of information.

- The original can be reconstructed exactly

Lossy — the compression process reduces the amount of information.

- -Only an approximation of the original can be reconstructed.

Categories of Compression Techniques

Entropy coding is lossless
Source coding and hybrid coding are lossy.

Coding techniques

Vector Quantization — a data stream is divided into blocks of n bytes(where n > 1). A predefined table contains a set of patterns is used to code the data blocks.

LZW —a general compression algorithm capable of working on almost any type of data. It builds a data dictionary of data occurring in an uncompressed data stream. Patterns of data are identified and are matched to entries in the dictionary. When a match is found the code of the entry is output.

Since the code is shorter than the data pattern, compression is achieved. The popular zip application used this method to compress files.

Differential coding — (also know as prediction or relative coding) The most known coding of this kind is DPCM (Differential Pulse Code Modulation). This method encodes the difference between the consecutive samples instead of the sample values. For example,

PCM 215 218 210 212 208 . . .

DPCM 215 3 -8 2 -4 . . .

DM (Delta Modulation) is a modification of DPCM. The difference is coded with a single bit.

Huffman coding

The principle of Huffman coding is to assign shorter code for symbol that has higher probability of occurring in the data stream.

The length of the Huffman code is optimal.

A Huffman code tree is created using the following procedures

- Two characters with the lowest probabilities are combined to form a binary tree.
- The two entries in the probability table is replaced by a new entry whose value is the sum of the probabilities of the two characters.
- Repeat the two steps above
- Assign 0 to be left branches and 1 to the right branches of the binary tree.
- The Huffman code of each character can be read from the tree starting from the root.

JPEG

- JPEG (stands for Joint Photographic Experts Group) is a joint ISO and CCITT (*ComitéConsultatif International Téléphonique et Télégraphique*),
- working group for developing standards for compressing still images
- The JPEG image compression standard became an international standard in 1992
- JPEG can be applied to colour or grayscale images.

Our eyes and ears cannot distinguish subtle changes. In such cases, we can use a lossy data compression method. These methods are cheaper—they take less time and space when it comes to sending millions of bits per second for images and video. Several methods have been developed using lossy compression techniques.

JPEG (Joint Photographic Experts Group) encoding is used to compress pictures and graphics, MPEG (Moving Picture Experts Group) encoding is used to compress video, and MP3 (MPEG audio layer 3) for audio compression.

Image compression – JPEG encoding

An image can be represented by a two-dimensional array (table) of picture elements (pixels).

A grayscale picture of 307,200 pixels is represented by 2,457,600 bits, and a color picture is represented by 7,372,800 bits.

In JPEG, a grayscale picture is divided into blocks of 8 × 8 pixel blocks to decrease the number of calculations because, as we will see shortly, the number of mathematical operations for each picture is the square of the number of units.

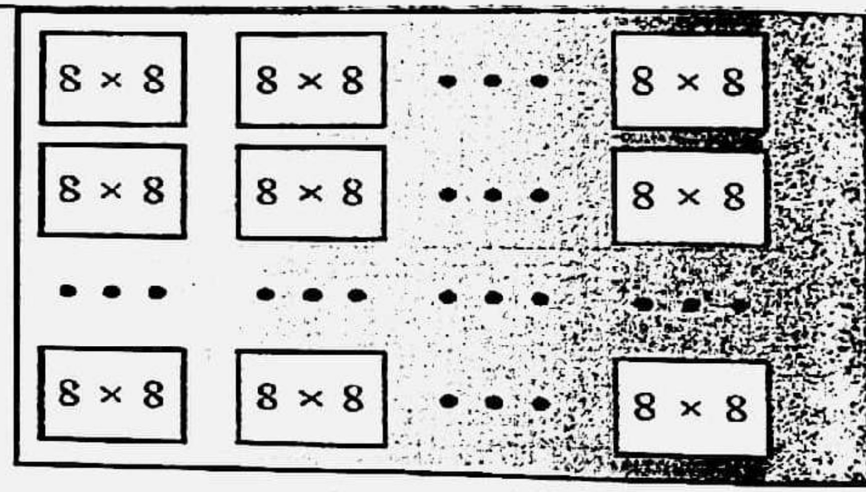


Figure 2: JPEG grayscale example, 640 × 480 pixels

The whole idea of JPEG is to change the picture into a linear (vector) set of numbers that reveals the redundancies. The redundancies (lack of changes) can then be removed using one of the lossless compression methods we studied previously. A simplified version of the process is shown in Figure 3.

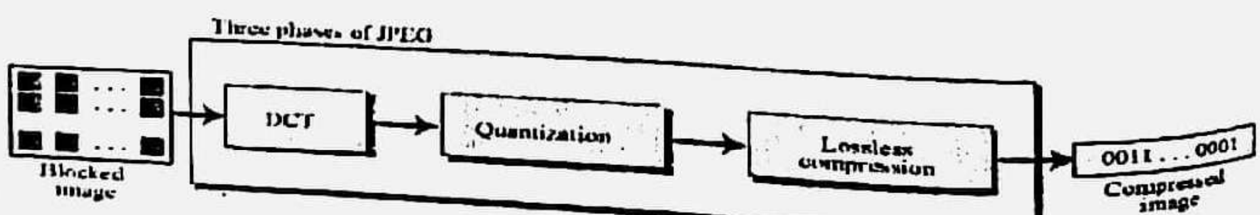


Figure 3 The JPEG compression process

Video compression – MPEG encoding

The Moving Picture Experts Group (MPEG) method is used to compress video. In principle, a motion picture is a rapid sequence of a set of frames in which each frame is a picture.

In other words, a frame is a spatial combination of pixels, and a video is a temporal combination of frames that are sent one after another.

Compressing video, then, means spatailly compressing each frame and temporally compressing a set of frames.

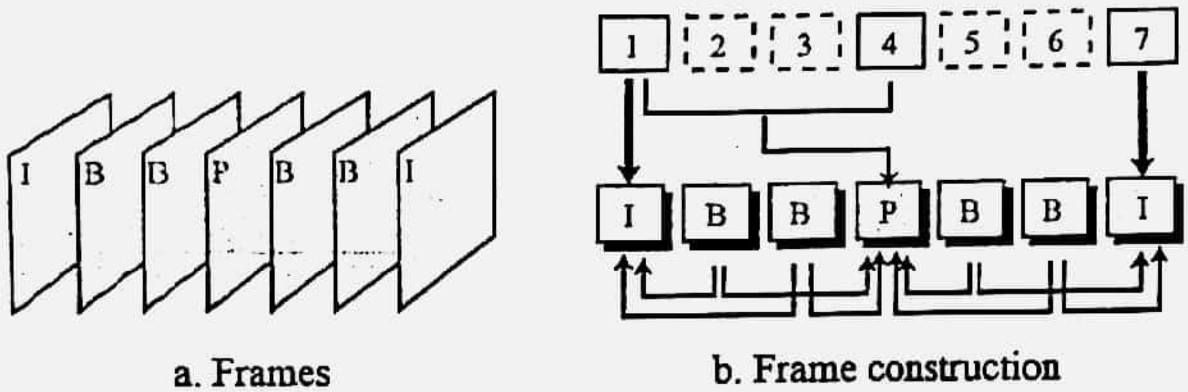


Figure 4 MPEG frames

Audio compression

Audio compression can be used for speech or music. For speech we need to compress a 64 kHz digitized signal, while for music we need to compress a 1.411 MHz signal. Two categories of techniques are used for audio compression: predictive encoding and perceptual encoding.

By changing appropriate parameters, the user can select

– the quality of the reproduced image

– compression processing time

– the size of the compressed image

The JPEG standard have three levels of definition as follows:
Baseline system — must reasonably decompress colour images, maintain a high compression ratio, and handle from 4bits/pixel to 16bits/pixel.
Extended system covers the various encoding aspects such as variable length encoding, progressive encoding, and hierarchical mode of encoding.

Special lossless function— ensures that at the resolution at which the image is compressed, decompression results in no loss of any detail the was in the original image.

JPEG — Preparation

- A source image consists of at least one and at most 255 planes.
- Each plane Ci may have different number ofpixels in the horizontal (Xi) and vertical (Yi) dimension.
- The resolution of the individual plane may be different.
- Each pixel is represented by a number of bits p where $2 \leq p \leq 12$.

The meaning of the value in these planes is not specified in the standard.

The image is divided into 8 X 8 blocks.

MPEG

- MPEG (stands for Moving Picture Experts Group) is also a joint ISO and CCITT working group for developing standards for compressing still images
- The MPEG video compression standard became an international standard in 1993
- MPEG uses technology defined in other standards, such as JPEG and H.261.
- It defines a basic data rate of 1.2Mbits/sec
- It is suitable for symmetric as well asymmetric compression It follows the reference scheme that consists of four stages of processing:
  - Preparation
  2) Processing
  3) Quantization
  4) Entropy Encoding
- In the preparation stage, unlike JPEG, MPEG defines the format of the images Each image consists of three components — YUV
- The luminance component has twice as many samples in the horizontal and vertical
- axes as the other two components (known as colour sub-sampling
- The resolution of the luminance component should not exceed 768 pixels
- for each component, a pixel is coded with eight bits

How MPEG encode the video stream?

In order to achieve higher compression ratio, MPEG uses the fact the image on consecutive frames differ relatively small. It uses a temporal prediction technique to encode the frame so that the storage requirement is greatly reduced.

Common MPEG data stream consists of four kinds of frames:

I-frame (Intra-frame) — it is a self-contained frame, and it is coded without reference to any other frames.

P-frame (Predictive-coded frame) — It is coded using the predictive technique with reference

pixels/line	line/frame	frames/sec
352	288	30
720	576	30
1920	1152	60

to the previous I-frame and/or previous P-frame.

B-frame (Bi-directionally predictive coded frame) — It requires information of the previous and following I_ and P_frames for encoding and decoding.

D-frame (DC-coded frame) Only the lowest frequency component of image is encoded. It is used in fast forward or fast rewind.

MPEG-2

MPEG-2 is a newer video encoding standard which builds on MPEG-1

It supports higher video quality and higher data rate (up to 80 Mbits/sec)

It supports several resolutions:

Summary

Compression methods — lossless vs. lossy

Entropy coding — run-length encoding, Huffman encoding

Source coding — prediction (DPCM, DM), transformation (DCT)

hybrid coding — JPEG, MPEG

Unit 6: User interface design

Objectives

- To suggest some general design principles for user interface design
- To explain different interaction styles and their use
- To explain when to use graphical and textual information presentation
- To explain the principal activities in the user interface design process
- To introduce usability attributes and approaches to system evaluation

The user interfaces

- User interfaces should be designed to match the skills, experience and expectations of its anticipated users.
- System users often judge a system by its interface rather than its functionality.
- A poorly designed interface can cause a user to make catastrophic errors.
- Poor user interface design is the reason why so many software systems are never used

Human factors in interface design

- **Limited short-term memory**
- People can instantaneously remember about 7 items of information. If you present more than this, they are more liable to make mistakes.
- **People make mistakes**
- When people make mistakes and systems go wrong, inappropriate alarms and messages can increase stress and hence the likelihood of more mistakes.
- **People are different**
- People have a wide range of physical capabilities. Designers should not just design for their own capabilities.
- **People have different interaction preferences**
- Some like pictures, some like text.

UI design principles

- UI design must take account of the needs, experience and capabilities of the system users.
- Designers should be aware of people's physical and mental limitations (e.g. limited short-term memory) and should recognize that people make mistakes.
- UI design principles underlie interface designs although not all principles are applicable to all designs.

Design principles

➢ **User familiarity**

The interface should be based on user-oriented terms and concepts rather than computer concepts. For example, an office system should use concepts such as letters, documents, folders etc. rather than directories, file identifiers, etc.

➢ **Consistency**

The system should display an appropriate level of consistency. Commands and menus should have the same format, command punctuation should be similar, etc.

➢ **Minimal surprise**

If a command operates in a known way, the user should be able to predict the operation of comparable commands

User-interface design principles

Principle	Description
User familiarity	The interface should use terms and concepts which are drawn from the experience of the people who will make most use of the system.
Consistency	The interface should be consistent in that, wherever possible, comparable operations should be activated in the same way.
Minimal surprise	Users should never be surprised by the behavior of a system.
Recoverability	The interface should include mechanisms to allow users to recover from errors.
User guidance	The interface should provide meaningful feedback when errors occur and provide context-sensitive user help facilities.

User diversity	The interface should provide appropriate interaction facilities for different types of system user.

Design issues in UIs

- ❖ Two problems must be addressed in interactive systems design
- ❖ How should information from the user be provided to the computer system?
- ❖ How should information from the computer system be presented to the user?
- ❖ User interaction and information presentation may be integrated through a coherent framework such as a user interface metaphor.

Interaction styles

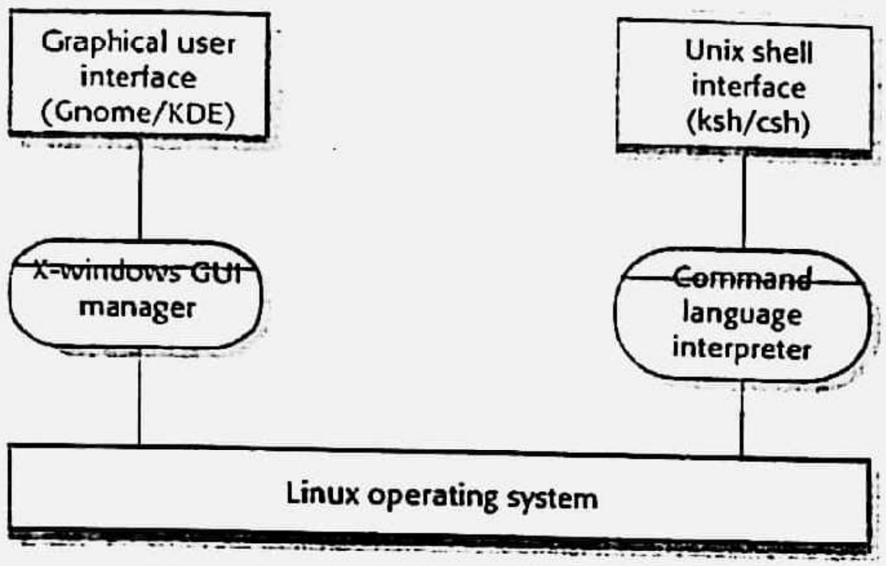
- Direct manipulation
- Menu selection
- Form fill-in
- Command language
- Natural language

Interaction style	Main advantages	Main disadvantages	Application examples
Direct manipulation	Fast and intuitive interaction Easy to learn	May be hard to implement. Only suitable where there is a visual metaphor for tasks and objects.	Video games CAD systems
Menu selection	Avoids user error Little typing required	Slow for experienced users. Can become complex if many menu options.	Most general-purpose systems
Form fill-in	Simple data entry Easy to learn Checkable	Takes up a lot of screen space. Causes problems where user options do not match the form fields.	Stock control, Personal loan processing
Command language	Powerful and flexible	Hard to learn. Poor error management.	Operating systems, Command and control systems
Natural language	Accessible to casual users Easily extended	Requires more typing. Natural language understanding systems are unreliable.	Information retrieval systems

LIBSYS interaction

- ➢ Document search
- ➢ Users need to be able to use the search facilities to find the documents that they need
- ➢ Document request
- ➢ Users request that a document be delivered to their machine or to a server for printing

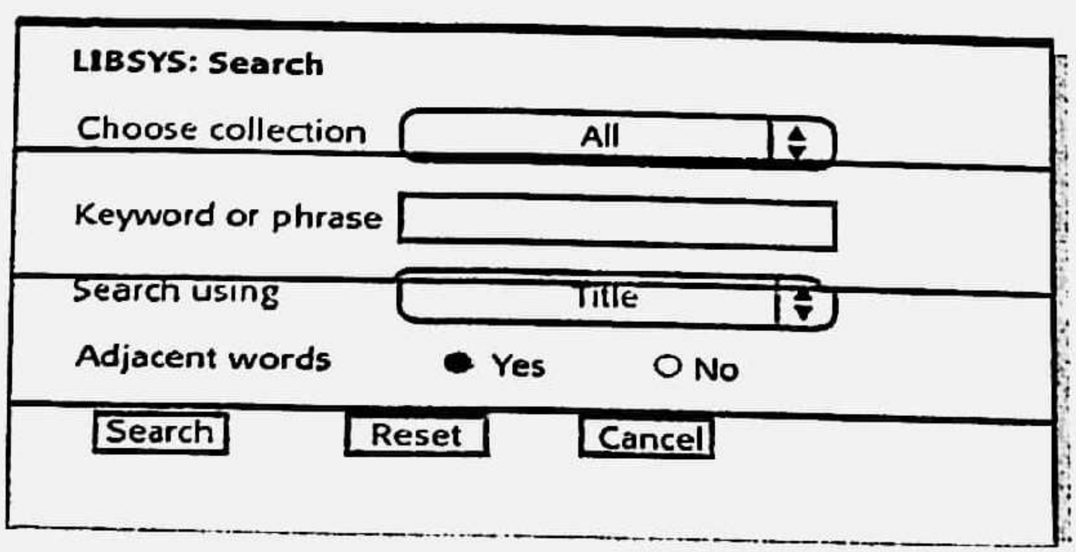
Multiple userinterfaces



Web-basedinterfaces

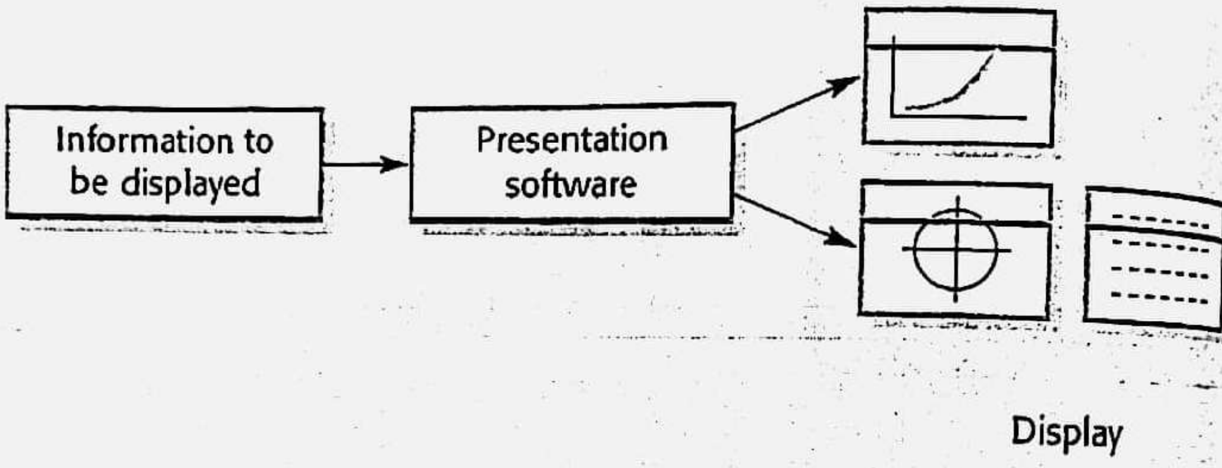
- Many web-based systems have interfaces based on web forms.
- Form field can be menus, free text input, radio buttons, etc.
- In the LIBSYS example, users make a choice of where to search from a menu and type the search phrase into a free text field.

LIBSYS searchform



Information presentation

- Information presentation is concerned with presenting system information to system users.
- The information may be presented directly (e.g. text in a word processor) or may be transformed in some way for presentation (e.g. in some graphical form).
- The Model-View-Controller approach is a way of supporting multiple presentations of data.



Information display factors

- Is the user interested in precise information or data relationships?
- How quickly do information values change? Must the change be indicated immediately?
- Must the user take some action in response to a change?
- Is there a direct manipulation interface?
- Is the information textual or numeric? Are relative values important?

Analogue or digital presentation?

- Digital presentation
- Compact - takes up little screen space;
- Precise values can be communicated.
- Analogue presentation
- Easier to get an 'at a glance' impression of a value;
- Possible to show relative values;
- Easier to see exceptional data values.

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Model-view-controller

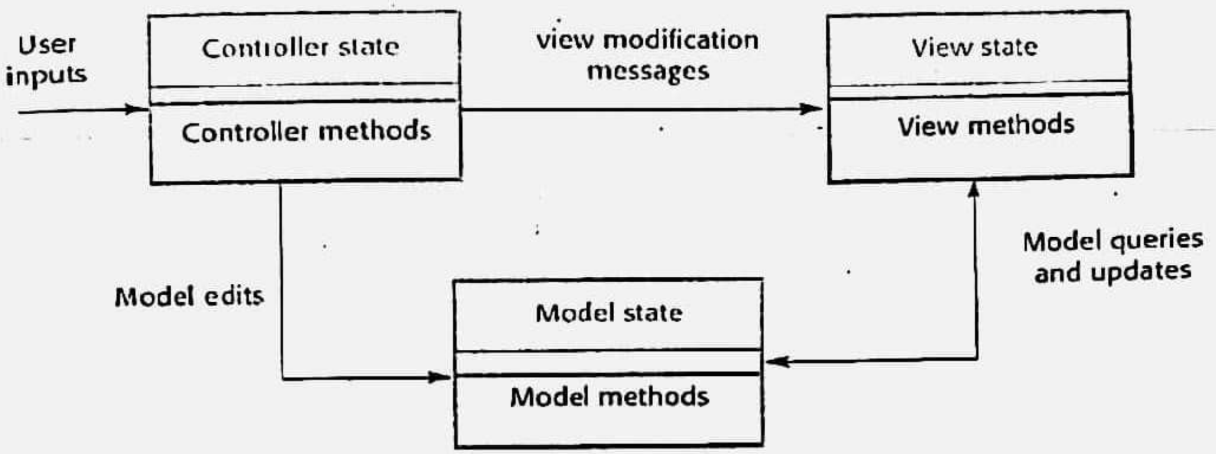


Fig: Model-view-controller

Alternative information presentations

Jan	Feb	Mar	April	May	June
2842	2851	3164	2789	1273	2835

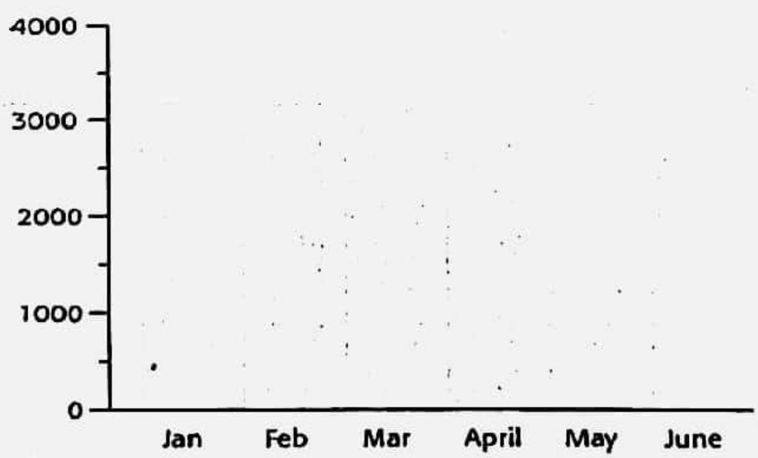
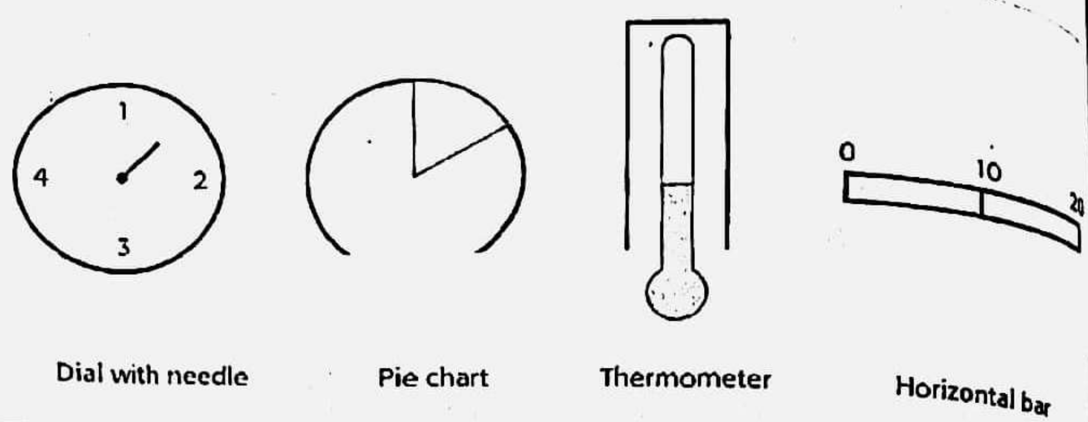
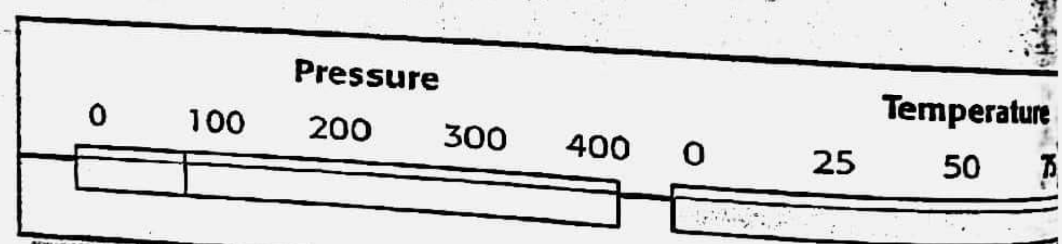


Fig: Alternative Information Presentation

Presentation Methods



Displaying relative values



Data visualisation

- Concerned with techniques for displaying large amounts of information.
- Visualization can reveal relationships between entities and trends in the data.
- Possible data visualizations are:
- Weather information collected from a number of sources;
- The state of a telephone network as a linked set of nodes;
- Chemical plant visualized by showing pressures and temperatures in a linked set of pipes;
- A model of a molecule displayed in 3 dimensions;
- Web pages displayed as a hyperbolic tree.

Colour displays

- Colour adds an extra dimension to an interface and can help the user understand complex information structures.
- Colour can be used to highlight exceptional events.
- Common mistakes in the use of colour in interface design include:
- The use of colour to communicate meaning;
- The over-use of colour in the display.

Colour use guidelines

- Limit the number of colours used and be conservative in their use.
- Use colour change to show a change in system status.
- Use colour coding to support the task that users are trying to perform.
- Use colour coding in a thoughtful and consistent way.
- Be careful about colour pairings.

Error messages

- Error message design is critically important.
- Poor error messages can mean that a user rejects rather than accepts a system.
- Messages should be polite, concise, consistent and constructive.
- The background and experience of users should be the determining factor in message design.

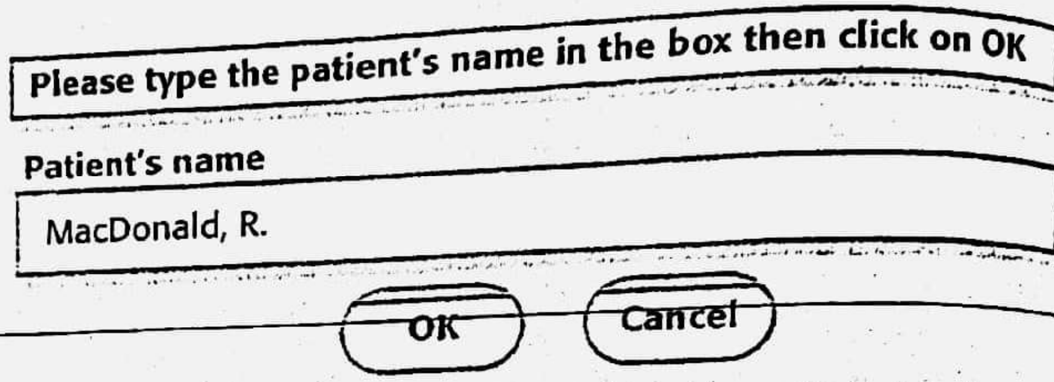
Design factors in message wording

Factor	Description
Context	Wherever possible, the messages generated by the system should reflect the current user context. As far as is possible, the system should be aware of what the user is doing and should generate messages that are relevant to their current activity.
Experience	As users become familiar with a s system, they become irritated by long, meaningful messages. However, beginners find it difficult to understand short terse statements of a problem. You should provide both types of message and allow the user to control message conciseness.
Skill level	Messages should be tailored to the user skills as well as their experience. Messages for the different classes of user may be ex pressed in different ways depending on the terminology that is familiar to the reader.
Style	Messages should be positive rather than negative. They should use the active rather than the passive mode of address. They should never be insulting or try to be funny.
Culture	Wherever possible, the designer of messages should be familiar with the culture of the country where the system is sold. There are distinct cultural differences between Europe, Asia and

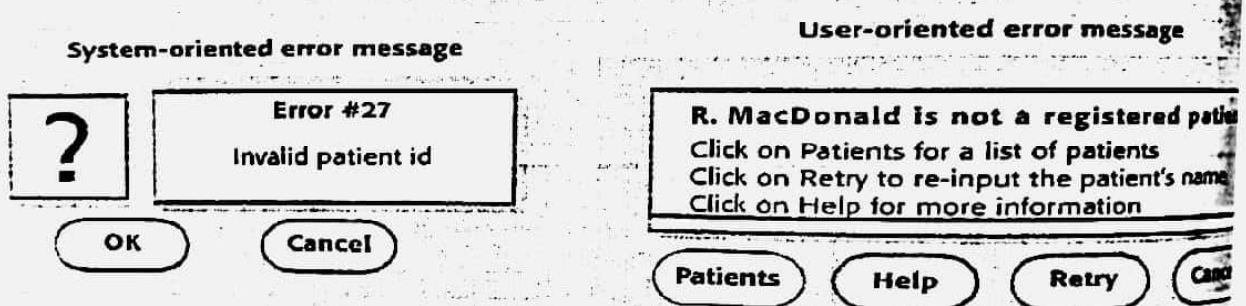
	America. A suitable message for one culture might be unacceptable in another.

User error

- Assume that a nurse misspells the name of a patient whose records he is trying to retrieve.



Good and Bad messages



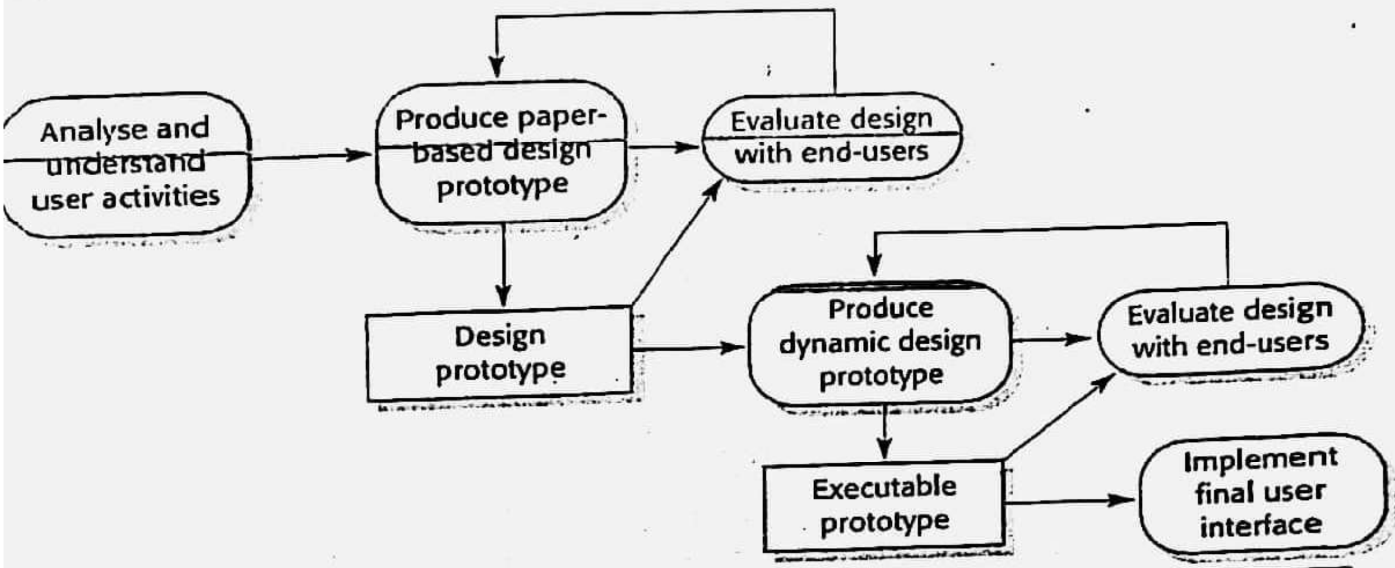
The UI design process

- UI design is an iterative process involving close liaisons between users and designers.
- The 3 core activities in this process are:

  User analysis. Understand what the users will do with the system;

  System prototyping. Develop a series of prototypes for experiment;

  Interface evaluation. Experiment with these prototypes with users.



User analysis

- If you don't understand what the users want to do with a system, you have no realistic prospect of designing an effective interface.
- User analyses have to be described in terms that users and other designers can understand.
- Scenarios where you describe typical episodes of use, are one way of describing these analyses.

User interaction scenario

Jane is a student of Religious Studies and is working on an essay on Indian architecture and how it has been influenced by religious practices. To help her understand this, she would like to access some pictures of details on notable buildings but can't find anything in her local library.
She approaches the subject librarian to discuss her needs and he suggests some search terms that might be used. He also suggests some libraries in New Delhi and London that might have this material so they log on to the library catalogues and do some searching using these terms. They find some source material and place a request for photocopies of the pictures with architectural detail to be posted directly to Jane.

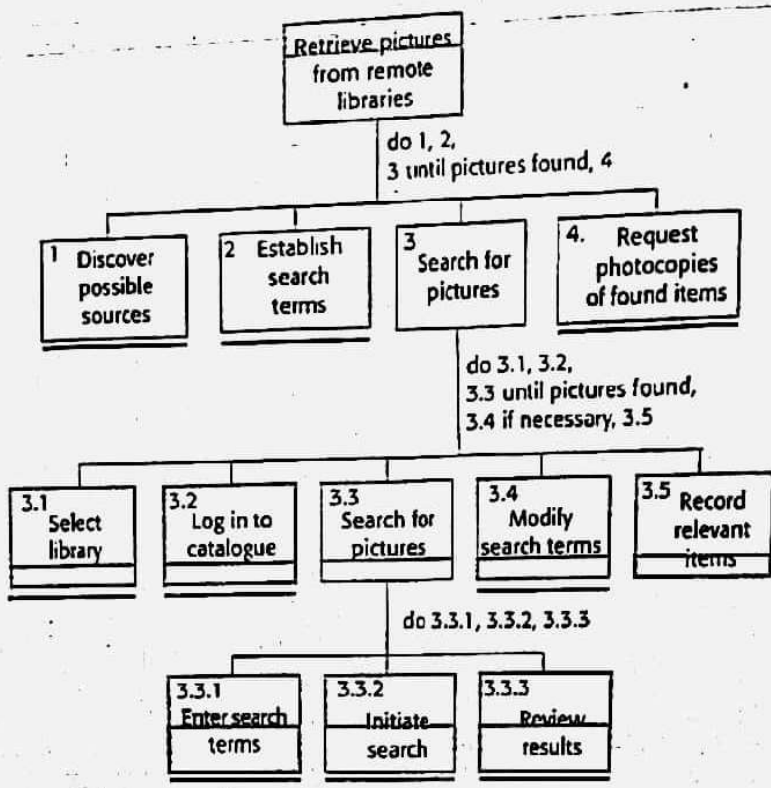
Requirements from the scenario

- Users may not be aware of appropriate search terms so need a way of helping them choose terms.
- Users have to be able to select collections to search.
- Users need to be able to carry out searches and request copies of relevant material.

Analysis techniques

- Task analysis
- Models the steps involved in completing a task.
- Interviewing and questionnaires
- Asks the users about the work they do.
- Ethnography
- Observes the user at work.

Hierarchical task analysis



Interviewing

- Design semi-structured interviews based on open-ended questions.
- Users can then provide information that they think is essential; not just information th have thought of collecting.
- Group interviews or focus groups allow users to discuss with each other what they do.

Ethnography

- Involves an external observer watching users at work and questioning them in an unsc way about their work.
- Valuable because many user tasks are intuitive and they find these very difficult to des and explain.
- Also helps understand the role of social and organizational influences on work.

Ethnographic records

Air traffic control involves a number of controls 'suites' where the suites controlling adjacent of airspace are physically located next to each other. Flights in a sector are represented by strips that are fitted into wooden racks in an order that reflects their position in the sector. If th not enough slots in the rack (i.e. when the airspace is very busy), controllers spread the strips the desk in front of the rack.

When we were observing controllers, we noticed that controllers regularly glanced at the str in the adjacent sector. We pointed this out to them and asked them why they did this. They that, if the adjacent controller has strips on their desk, then this meant that they would have flights entering their sector. They therefore tried to increase the speed of aircraft in the se 'clear space' for the incoming aircraft.

Insights from ethnography

- Controllers had to see all flights in a sector. Therefore, scrolling displays where flights disappeared off the top or bottom of the display should be avoided.
- The interface had to have some way of telling controllers how many flights were in adjacent sectors so that they could plan their workload.

User interface prototyping

- The aim of prototyping is to allow users to gain direct experience with the interface.
- Without such direct experience, it is impossible to judge the usability of an interface.
- Prototyping may be a two-stage process:
- Early in the process, paper prototypes may be used;
- The design is then refined and increasingly sophisticated automated prototypes are then developed.

Paper prototyping

- Work through scenarios using sketches of the interface.
- Use a storyboard to present a series of interactions with the system.
- Paper prototyping is an effective way of getting user reactions to a design proposal.

Prototyping techniques

- Script-driven prototyping
- Develop a set of scripts and screens using a tool such as Macromedia Director. When the user interacts with these, the screen changes to the next display.
- Visual programming
- Use a language designed for rapid development such as Visual Basic. See Chapter 17.
- Internet-based prototyping
- Use a web browser and associated scripts.

User interface evaluation

- Some evaluation of a user interface design should be carried out to assess its suitability. Full scale evaluation is very expensive and impractical for most systems.
- Ideally, an interface should be evaluated against a usability specification. However, it is rare for such specifications to be produced.

Usability attributes

Attribute	Description
Learnability	How long does it take a new user to become productive with the system?
Speed of operation	How well does the system response match the user's work practice?
Robustness	How tolerant is the system of user error? Recoverability
Recoverability	How good is the system at recovering from user errors?

Scanned with CamScanner

Simple evaluation techniques

- Questionnaires for user feedback.
- Video recording of system use and subsequent tape evaluation.
- Instrumentation of code to collect information about facility use and user errors.
- The provision of code in the software to collect on-line user feedback.

Key points

- User interface design principles should help guide the design of user interfaces.
- Interaction styles include direct manipulation, menu systems form fill-in, comma languages and natural language.
- Graphical displays should be used to present trends and approximate values. Digi displays when precision is required.
- Colour should be used sparingly and consistently.

Key points

- The user interface design process involves user analysis, system prototyping a prototype evaluation.
- The aim of user analysis is to sensitive designers to the ways in which users actual work.
- UI prototyping should be a staged process with early paper prototypes used as a ba for automated prototypes of the interface.
- The goals of UI evaluation are to obtain feedback on how to improve the interfa design and to assess if the interface meets its usability requirements.

Unit 7: Abstraction for Programming

- The state of the art of programming
- Most of the current commercially available multimedia applications are implemented in procedure-oriented programming languages
- Application code is still highly dependent on hardware
- Change of multimedia devices still often requires re- implementation
- Common operating system extensions try to attack these problems
- Different programming possibilities for accessing and representing multimedia data

Overview of different abstraction levels

- Libraries
- System software
- Toolkits
- Higher Programming languages
- Object-oriented approaches

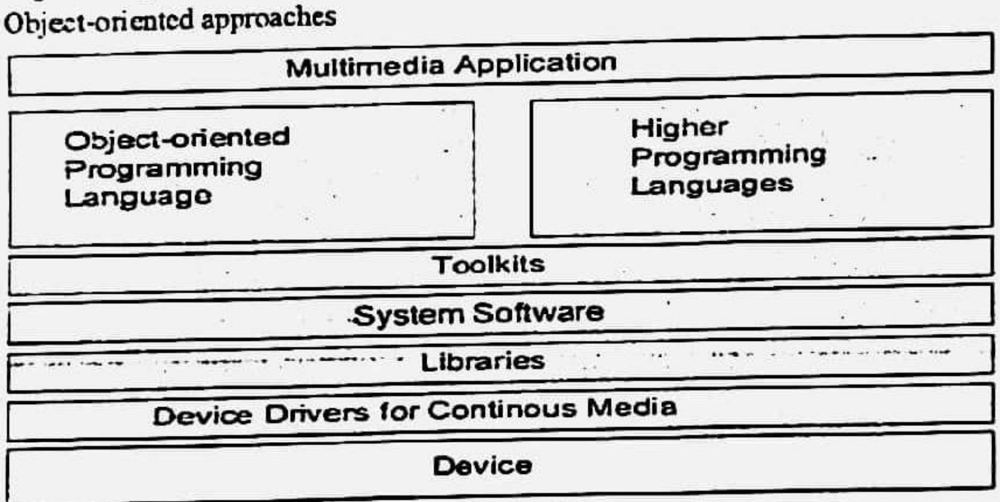


Figure: Abstraction for Programming

Abstractions from Multimedia Hardware

Strong hardware dependency may cause problems with:

- Portability
- Reusability
- Coding efficiency

Abstraction Levels

- Common operating system extensions try to solve this problem
- Different programming possibilities for accessing and representing multimedia data

Libraries

- Processing of continuous media based on functions embedded in libraries
- Libraries differ in their degree of abstraction

Libraries Open GL

2D and 3D graphics API developed by Silicon Graphics

Basic idea: "write applications once, deploy across many platforms":

- PCs
- Workstations
- Super Computers

Benefits:

- Stable
- Reliable and Portable
- Evolving
- Scalable (Features like Zoom, Rectangle handling ...)
- Well documented and easy to use

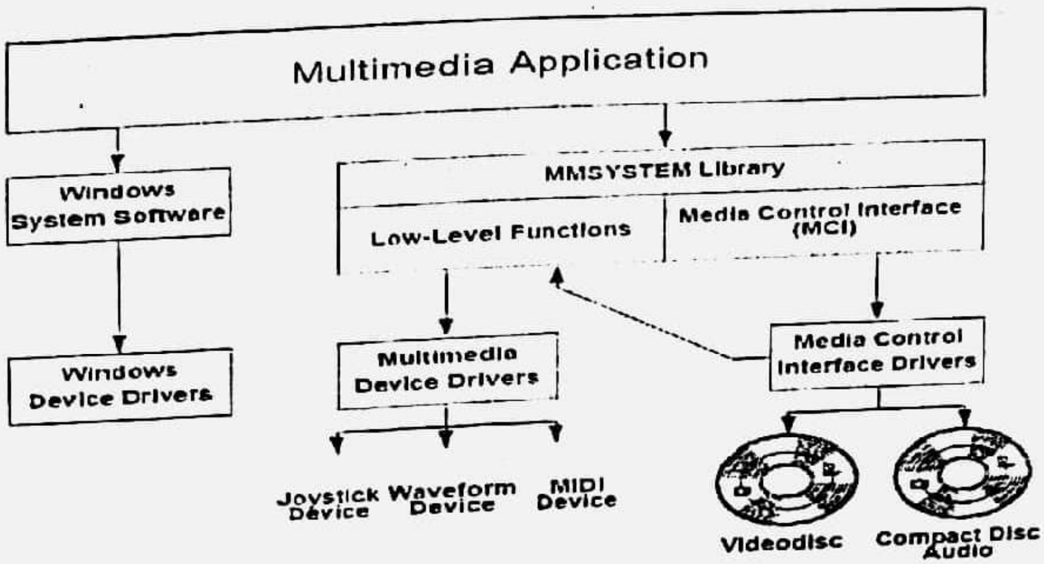
• Integrated with:

- Windows 95/NT/2000/XP
- UNIX X Window System

System Software

- Device access becomes part of the operating system:
- Data as *time capsules (file extensions)*
  - Each Logical Data Unit (LDU) carries in its time capsule its data type, actual value and valid life span
  - Useful concept for video, where each frame has a valid life span of 40ms (rat of read access during a normal presentation)
  - Presentation rate is changed for VCR (Video Cassette Recorder) functions lik fast forward, slow forward or fast rewind by Changing the presentation life span of a LDU
  - -Skipping of LDUs or repetition of LDUs
- **Data as *streams***
  - a stream denotes the continuous flow of audio and video data between a sourc and a sink
  - Prior to the flow the stream is established equivalent to the setup of a connection in a networked environment

System Software: Windows Media Control Interface (MCI):



- MMSYSTEM library for extensibility and device independence

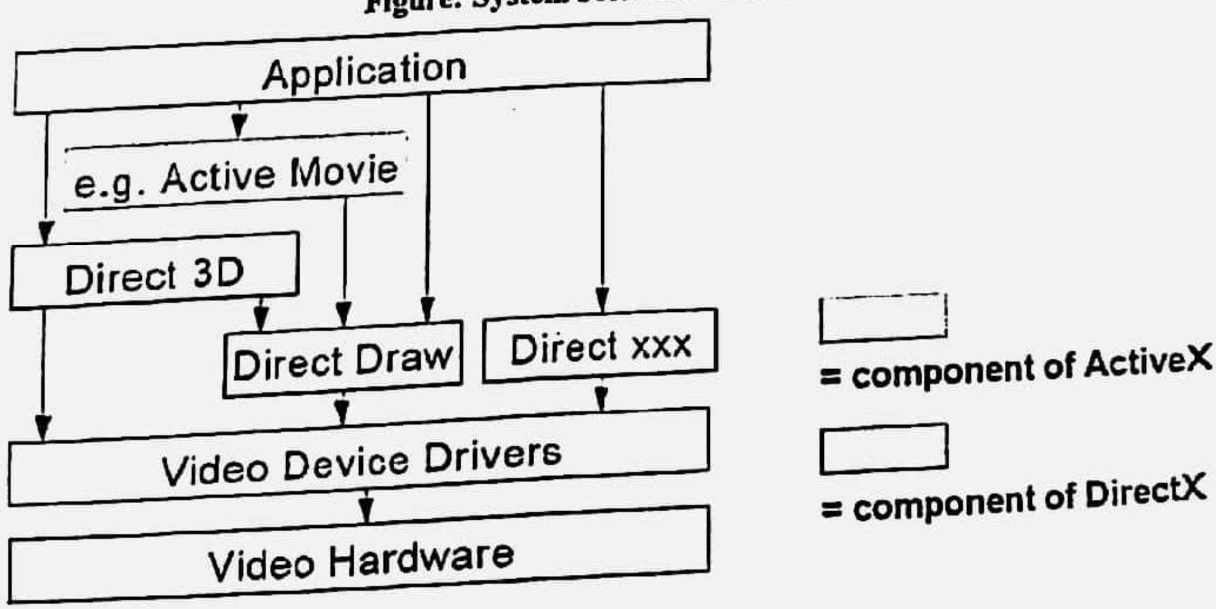
System Software - DirectX

- Low-level APIs and libraries for high-performance applications
- Especially games - formerly known as the "Game SDK"
- Direct access to hardware services
- E.g. audio & video cards, hardware accelerators

"DirectX" = "direct access"

- Strong relationship/interaction with ActiveX/DCOM

Figure: System Software - DirectX



Components:

- DirectDraw - 2-dimensional graphics capabilities
- Direct3D - extensively functional 3D graphics programming API
- DirectSound - (3D) sound, mixing and playback of multiple streams
- DirectPlay - for network multiplayer game development
- DirectInput - input from various peripherals, e.g. joysticks, data gloves

Implementation Strategy:

- Hardware Abstraction Layer (HAL)
- Hardware Emulation Layer (HEL)
- Media Layer (for aggregated "high level" functionality)
    - Animations
    - Media streaming
    - Synchronization

Toolkits

- Simpler approach than the system software interface from the user's point of view are toolkits:
- Abstract from the actual physical layer
- Allow a uniform interface for communication with all different devices of continuous media
- Introduce the client-server paradigm
- Can be embedded into programming languages or object-oriented environments.

Higher Programming Language:

Media as data types:

- Definition of appropriate data types (e.g. for video and audio)
- Smallest unit can be a LDU
- Example of merging a text and a motion picture:

Media as files:

instead of considering continuous media as data types they can be considered as files:

```
file_h1 = open(MICROPHONE_1,...)
file_h2 = open(MICROPHONE_2,...)
file_h3 = open(SPEAKER, ...)
...
read(file_h1)
read(file_h2)

mix(file_h3, file_h1, file_h2)
activate(file_h1, file_h2, file_h3)
..
```


Scanned with CamScanner

deactivate(file_h1, file_h2, file_h3)

...

rc1 = close(file_h1)
rc2 = close(file_h2)
rc3 = close(file_h3)

Programming Language Requirements

- The high-level language should support parallel processing, because the processing of continuous data is controlled by the language through pure asynchronous instructions an integral part of a program through the identification of media
- Different processes must be able to communicate through an inter- process communication mechanism, which must be able to:
- Understand a priori and/or implicitly specified time requirements (QoS parameters or extracted from the data type)
- Transmit the continuous data according to the requirements
- Initiate the processing of the received continuous process on time

Object-Oriented Approaches

Basic ideas of object-oriented programming is data encapsulation in connection with class and object definitions

- Abstract Type Definition (definition of data types through abstract interfaces)
- Class (implementation of a abstract data type)
- Object (instance of a class)
- Other important properties of object-oriented systems are:
  - Inheritance
  - Polymorphism
- Devices as classes: devices are assigned to objects which represent their behavior and interface

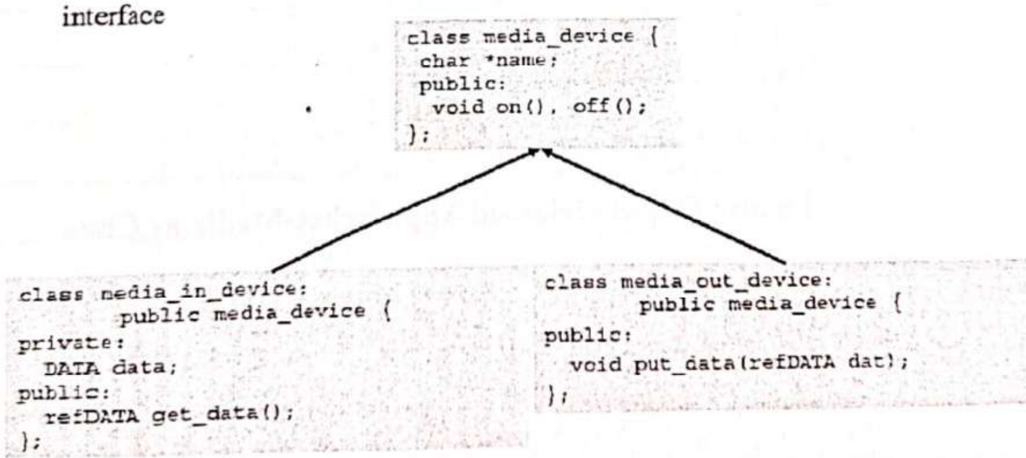


Figure: Devices as classes

Processing units as classes:

Three main objects:

- Source objects
- Destination objects
- Combined source-destination objects allow the creation of data flow paths through connection of objects

Multimedia object

- Basic Multimedia Classes (BMCs) / Basic Multimedia Objects (BMOs)
- Compound Multimedia Classes (CMCs) / Compound Multimedia Objects (CMO), which are compound of BMCs / BMOs and other CMCs/CMOs
- BMOs and CMOs can be distributed over different computer nodes

Media as classes:

- Media Class Hierarchies define hierarchical relations for different media
- Different class hierarchies are better suited for different applications

```
Medium
 Acoustic_Medium
  Music
   Opus
    Score
     Audio_Block
      Sample_Value
  Speech
   ...
  ...
 Opitcal_Medium
  Video
   Video_Scene
```

```
Video
 Video_Scene
   Image
     Image_Segment
       Pixel
     Line
       Pixel
     Column
       Pixel
Animation
 ...
Text
 ...
```

Figure: Object-Oriented Approaches-Media as Class